Algebra Review: Unit 4 — Graphing Inequalities on the Coordinate Plane

Answer Key

Station 1: Testing Solutions of Linear Inequalities

1. Yes $3 > 1$ *true*
2. No $2 \leq 1$ *false*
3. Yes $5 \geq 5$ *true*
4. No $12 < 12$ *false*
5. Yes $4 \geq 4$ *true*
6. No $-3 < -3$ *false*
7. Yes $7 > 5$ *true*
8. $(3, -1)$ only $0 > 0$ *F;* $-1 > -3$ *T;* $2 > 2$ *F*
9. $(0, 0)$ and $(2, 1)$ $0 < 2$ *T;* $1 < 4$ *T;* $4 < 1$ *F*
10. $y \leq 4$

Station 2: Boundary Lines and Shading

1. $y = 3x - 2$; dashed; shade above
2. $y = -x + 5$; solid; shade below
3. $y = \frac{1}{2}x$; solid; shade above
4. $y = -4$; dashed; shade below
5. $x = 3$; solid; shade to the right
6. $x = -1$; dashed; shade to the left
7. $y = -2x + 6$; dashed; shade above $y > -2x + 6$
8. $y = \frac{3}{2}x - 4$; solid; shade above *dividing by* -2 *flips the sign:* $y \geq \frac{3}{2}x - 4$
9. $y = -\frac{1}{4}x + 2$; dashed; shade below
10. $y = -x - 2$; solid; shade below $y \leq -x - 2$

Station 3: Writing Inequalities from Graphs and Situations

1. $y \leq 2x + 1$
2. $y > -x + 3$
3. $y > \frac{3}{2}x - 2$
4. $y \leq -2x + 4$ *slope* $= \frac{0 - 4}{2 - 0} = -2$
5. $y < \frac{1}{2}x$
6. $x \geq -2$
7. $y \leq 5$
8. $y > 2x - 1$ *line:* $y = 2x - 1$*; origin:* $0 > -1$
9. $y \geq \frac{3}{2}x + 3$ *origin:* $0 < 3$ *is below, so shade above*
10. $2x + 5y \leq 60$; $20 + 40 = 60 \leq 60$, so yes

Station 4: Systems of Linear Inequalities

1. Yes $3 > 2$ *and* $3 \leq 4$
2. No $1 \geq 1$ *true, but* $1 < 1$ *false*
3. Yes $4 \leq 5$ *and* $4 > 2$
4. Yes $0 > -2$ *and* $0 < 3$
5. $(2, 3)$
6. $(3, 5)$
7. $(0, 0),\ (6, 0),\ (0, 6)$
8. $(1, 0),\ (4, 0),\ (4, 3),\ (1, 3)$; area 9
9. No solutions *the lines are parallel; a point cannot be above the higher line and below the lower line*
10. $15x + 10y \geq 120,\ x + y \leq 15,\ x \geq 0,\ y \geq 0$; $(4, 6)$ works: $120 \geq 120$ and $10 \leq 15$